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Enhanced solubilization of solid organics and methane production by anaerobic digestion of swine manure under nano-bubble water addition

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ABSTRACT

Nano-bubble water (NBW) refers to water with a large number of nanoscale particle bubbles. The aim of this work was to study the mechanism of NBW addition into the anaerobic digestion (AD) of swine manure (SM). The results showed that the cumulative methane production from the NBW added reactor was 192–225 mL/g-VS and 19–39% higher than the control group (without NBW addition). Based on the analysis of soluble organics, NBW addition not only accelerated hydrolysis rates of proteins and carbohydrates, but also enhanced the production of VFAs. Moreover, mechanism analysis reveals that NBW with higher spin–spin relaxation time and absolute value of zeta potential might promote enzyme activity and the hydrolysis of organic solids. Simultaneously, the electron transport system activity of the methanogenic communities and electric conductivity were enhanced by NBW addition. This work implies that NBW addition is promising for enhancing AD for enhancement of methane production.

1. Introduction

Globally, a large amount of livestock manure from animal husbandry needs to be scientifically and effectively processed. The annual livestock waste produced in Japan is about 80 million tons, of which swine manure (SM) is as high as 21 million tons (MAFF, 2018). If these huge amounts of SM are not treated properly, not only the biomass resources are wasted, but also the surrounding environments are polluted. Composting is a widely applied method to handle livestock manure since it is capable of stabilizing organic matters, increasing the manure fertility and sterilizing weed seeds and pathogens (Yuan et al., 2017). However, composting has some disadvantages like excessive nutrients loss, greenhouse gases emission, and high eutrophication potential (Yuan et al., 2018).

Anaerobic digestion (AD) has been widely used in the treatment of SM. AD involves a series of biochemical processes, in which anaerobic microorganisms can degrade and stabilize organic matters and produce biogas and other valuable intermediates as well (Raposo et al., 2009). Compared with other traditional treatment methods such as composting, AD of SM has many advantages, such as water pollution reduction, generation of valuable bioenergy and fertilizer, odor control, pathogen inactivation (Atandi and Rahman, 2012). Although the AD process also produces CO_2, the carbon source is not derived from fossil fuel but from biomass, which is considered as carbon neutral (Kothari et al., 2014). Therefore, AD has become a hot topic in the treatment of SM. However, AD of SM also has some shortcomings, such as prone to inhibition, low hydrolysis rate, low biogas and methane production (Burton et al., 2003; Carrère et al., 2009).

Many methods can be used to increase methane production and to improve AD performance, such as chemical (González-Fernández et al., 2008), thermal (Wu et al., 2017), thermal steam explosion pretreatment (Ferreira et al., 2014), co-digestion (Dennehy et al., 2016), initial pH adjustment (Zhang et al., 2015), additives addition (Wang et al., 2017; Yang et al., 2018), or applying a proper inoculum ratio (González-Fernández and García-Encina, 2009). Although these methods show some improvement effects on AD performance, most of them have not been applied in industry. The main reason is most probably due to the consumption of chemicals and energy or the complexity of operations, partly counteracting its environmental-friendliness and cost-effectiveness. As known, water is critical in the AD process, which can facilitate the transfer of hydrolyzed products to the site where the bacteria can utilize them (Bollon et al., 2013). In this aspect, most studies have been focusing on water content, namely, AD performance under different total solids (TS) contents (dry or wet AD) (Li et al., 2015; Chen et al.,

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2015). Few of them could be found on the effects of water type and characteristics on AD performance. Nano-bubble water (NBW) refers to water with a large number of nanoscale particle bubbles. NBW is characterized by its reduced friction and consequently increasing the solubility of gases in liquids and producing free radicals (Liu et al., 2013). Because of its unique characteristics, NBW has been successfully applied in many fields, such as water purification (Kyzas et al., 2019; Agarwal et al., 2011), and medicine (Thakur et al., 2019). One recent preliminary trial found that adding NBW to the digester treating waste activated sludge could increase the methane production (Wang et al., 2019). Although this preliminary research did not conduct a specific study of the four stages of AD and did not analyze the changes in organic and inorganic substances in the AD process. This work is encouraging further research on the application effect of NBW in AD. However, little information is available on the effect of NBW in AD, not to mention the analysis of AD process and mechanisms involved under NBW addition.

The purpose of the present study was to investigate the effect of NBW on AD of SM, aiming at exploring a new method to improve AD performance rather than using conventional methods such as chemical addition, thermal treatment and co-digestion. In order to study the mechanisms of NBW in AD of SM, this study measured the spin–spin relaxation times (T_2) and zeta potential of NBW used in the AD process and analyzed the related key enzymes. To better understand the kinetics of AD process and then analyze AD performance, kinetic modeling was also applied in this study. To the best of our knowledge, this was the first systematic research on the effects of NBW addition on organics conversion, enzyme activity and the mechanisms involved in the AD process. In addition, this work was also the first study on the application of NBW in the AD process of livestock manure.

2. Materials and methods

2.1. Characterization of NBW

Three kinds of NBW were prepared, namely air-NBW (Air-NBW), hydrogen-NBW (H_2-NBW), and carbon dioxide-NBW (CO_2-NBW). The gases (Air, H_2 and CO_2) were introduced into deionized water (DW) through a nano-bubble generator (HACK FB11, Japan) for 20 min at 20 ± 2 °C to obtain three kinds of NBW based on a previous study (Wang et al., 2019). The zeta potential of NBW was detected by the Zeta Potential Analyzer (MALVERN, Nano ZS, United Kingdom). The electric conductivity was measured by an electric conductivity meter (AS-ONE CORPORATION, Waterproof portable conductivity meter AS710, China). The spin–spin relaxation times (T_2) were measured according to a previous research (Liu et al., 2013), in which a pulsed NMR (JNM-MU25A, JEOL, Japan) was used.

2.2. Batch AD experiments

2.2.1. Raw swine manure and inoculum

Raw SM was collected from a pig farm located in Tsukuba, Ibaraki, Japan. The pigs were fed with corn, soybean, bran, and so on. After manually sorting to remove large particles, such as small stones, plastics and small clods, the raw SM was mixed thoroughly to ensure its homogeneity and obtain the representative manure sample. The manure sample was then stored at 4 °C in a refrigerator before AD experiments. Main characteristics of the raw SM and seed AD sludge (inoculum) are shown in Table 1.

2.2.2. Batch AD tests

All batch AD experiments were performed in a series of 250 mL Scott Duran bottles, which were labeled as $R_{Control}$, R_{NBW}-Air, R_{NBW}-H_2 and R_{NBW}-CO_2 when the added water was DW, Air-NBW, H_2-NBW, and CO_2-NBW, respectively (Table 2). Each 250 mL bottle was loaded with 20 g RSM and 80 g inoculum. In this research, 100 mL NBW or DW was added at the beginning of the experiment. Two bottles only containing inoculum were also tested in this work. As pH adjustment before AD may have some impact on the microbial communities of raw SM, all the initial sample pH before AD experiments were not adjusted in this study. After being loaded with all the materials, the bottles were sealed using rubber stoppers. All the reactors were flushed with N_2 for 3 min and then put these reactors in an incubator (35 ± 2 °C) for 28 days based on the preliminary experiments. The bottles were shaken manually for 2 min once per day at a fixed time before gas production measurement. In this research, each AD test was conducted in duplicate and eight bottles were sacrificed for sampling and analysis every four days, four of which were parallel groups.

Table 1
Characteristics of raw swine manure and inoculum used in this study.

Parameters	Unit	Raw swine manure	inoculum
Total solids (TS)	%	25.58 ± 0.22	1.86 ± 0.06
Volatile solids (VS)	%	20.47 ± 0.29	1.35 ± 0.04
Total nitrogen (TN)	mg/g	7.51 ± 0.09	1.46 ± 0.07
Total ammonia nitrogen (TAN)	mg/g	2.82 ± 0.02	0.77 ± 0.05
Total alkalinity (TA, as $CaCO_3$)	mg/g	16.59 ± 0.37	ND
Total phosphorus (TP)	mg/g	5.76 ± 0.16	ND
Organic phosphorous (OP)	mg/g	0.96 ± 0.02	ND
Inorganic phosphorous (IP)	mg/g	4.79 ± 0.10	ND
Soluble total organic carbon (sTOC)	mg/g	10.39 ± 0.62	ND
C	%/TS	38.62 ± 0.46	45.42 ± 0.60
H	%/TS	5.70 ± 0.19	5.41 ± 0.26
N	%/TS	3.62 ± 0.09	3.78 ± 0.13
C/N	–	10.67	12.02
pH	–	6.77 ± 0.03	7.38 ± 0.04

ND (Not Determined).
Data in the table are expressed as Means ± SD.

Table 2
Experimental conditions for $R_{Control}$ and R_{NBW} in this study.

	R_{NBW} –CO_2	R_{NBW} –H_2	R_{NBW} -Air	$R_{Control}$
Digested sludge (g)	80	80	80	80
Raw swine manure (g) DW (g)	20	20	20	20
	0	0	0	100
H_2-NBW (g)	0	100	0	0
CO_2-NBW (g)	100	0	0	0
Air-NBW (g)	0	0	100	0

2.2.3. Other analytical methods

Total solids (TS), volatile solids (VS), total nitrogen (TN) and total ammonia nitrogen (TAN) were determined by the standard method (APHA, 2012). P was determined according to Standards, Measurements and Testing (SMT) protocol (García-Albacete et al., 2012). During the AD, the pH was determined by a pH meter (METTLER TOLEDO FE20, Switzerland). The C/N analysis was conducted using an elemental analyzer (PerkinElmer 2400 II CHN Elemental Analyzer, USA). Soluble carbohydrates concentration was measured by using the Phenol-sulfuric method. Soluble proteins concentration was determined with the Coomassie Brilliant Blue method. Before analysis, the sample was centrifuged, then filtered through a 0.22 μm membrane and the filtrate was used for soluble components quantification. As for volatile fatty acids (VFAs) analysis, phosphoric acid solution (3%) was added to the filtrate before injection to the gas chromatograph (GC-8A, SHIMADZU CO., Japan) equipped with flame ionization detector (FID) and Unisole F-200 30/60 column. In this determination, the temperatures of injector and column were 180 °C and 150 °C, respectively.

Biogas (CH_4 + CO_2 + H_2) production was determined using a gas tight 60 mL syringe, and the content of CH_4 was determined by a gas chromatograph (GC-8A/TCD, SHIMADZU CO., Japan) with N_2 as the

carrier gas. Net CH_4 production was obtained by subtracting the amount produced by the inoculum. Total alkalinity (TA) was determined according to a previous study (Sun et al., 2016). Soluble total organic carbon (sTOC), soluble inorganic carbon (sIC) and soluble total carbon (sTC) were determined by a TOC analyzer (TOC-VCSN, ASI-V autosampler, SHIMADZU CO., Japan). The electron transport system (ETS) activity was determined by INT method (Yin et al., 2005).

To explore the mechanisms involved in AD with the addition of NBW, the activities of six enzymes, including coenzyme F_{420}, dehydrogenase, protease, α-glucosidase, acid phosphatase (ACP) and alkaline phosphatase (ALP), were examined individually in the experiments. Coenzyme F_{420} and dehydrogenase were tested every 4 days during the AD. The other four enzymes were measured on day 4. Coenzyme F_{420} was measured by using the ultraviolet spectrophotometry method (Hutschemackers et al., 1982). Dehydrogenase activity was tested by using the TTC-Dehydrogenase method (Klapwijk et al., 1974). Protease, α-glucosidase, ALP and ACP were measured according to the literatures (Goel et al., 1998; Yuan et al., 2017). Briefly, protease was determined by using casein (10 mg/mL) as the substrate and 0.4 M trichloroacetic acid (TCA) was used to terminate the reaction. The p-nitrophenyl phosphate disodium salt (pnpP) of 0.1% was used as the substrate for the determination of ALP and ACP. The p-nitrophenyl-a-D-glucopyranoside (pnpG) of 0.1% was used as the substrate for the determination of α-glycosidase.

2.2.4. Calculation methods

VS reduction rate (%) after the AD process was calculated by Eq. (1)

$$\text{VS reduction rate}(\%) = \frac{[VS]_{initial} - [VS]_{final}}{[VS]_{initial}} \times 100 \quad (1)$$

The concentration of sTOC was calculated by Eq. (2)

$$sTOC = sTC - sIC \quad (2)$$

2.3. Kinetic models

Two models were used to study the methane production process. The modified Gompertz model (Eq. (3)) was used to calculate maximum CH_4 production potential, maximum CH_4 production rate and lag phase. The first-order model (Eq. (4)) was introduced to calculate the hydrolysis constant.

$$P = P_{\max} \exp\left\{-\exp\left[\frac{R_{\max} e}{P_{\max}}(\lambda - t) + 1\right]\right\} \quad (3)$$

$$P = P_{max}(1 - exp(-k_H \cdot t)) \quad (4)$$

where P is the cumulative CH_4 production (mL CH_4/g-VS), $P_{\max}$ is the maximum CH_4 production potential (mL CH_4/g-VS), $R_{\max}$ is the maximum CH_4 production rate (mL CH4/g-VS·d), λ is the lag phase (d). k_H is the first-order hydrolysis rate constant (d^{-1}). t is the duration of the AD process (d).

2.4. Statistical analysis

For AD experiments and characterization tests for NBW, mean values from six repetitions of analysis were reported based on three parallel trials by using Microsoft Office Excel 2016. One-way analysis of variance (ANOVA) was also applied for statistical analysis. Significant difference was set at $p < 0.05$. Multiple comparison analysis was used paired t-test with Bonferoni to find the most effective NBW for methane production. Significant difference was set at $p < 0.0083$.

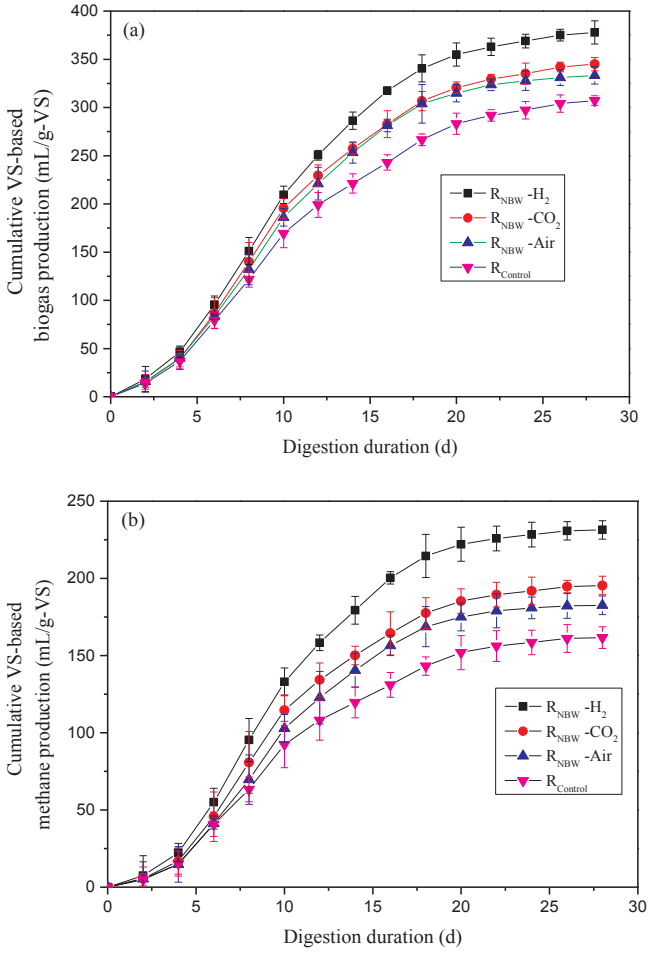


Fig.1. (a) Cumulative VS-based biogas production. (b) Cumulative VS-based methane production.

3. Results and discussion

3.1. Biogas production and methanogenesis performance

Biogas and methane productions were evaluated in the batch tests to study the effect of NBW addition on AD of SM. As shown in Fig. 1, the cumulative biogas and methane productions from R_{NBW} were higher than those from $R_{Control}$. For the cumulative biogas production (Fig. 1a), R_{NBW}-H_2, R_{NBW}-CO_2 and R_{NBW}-Air achieved 378, 345 and 333 mL/g-VS, respectively, corresponding to 31.5%, 20.1% and 16.1% increase when compared with $R_{Control}$. In terms of cumulative methane production (Fig. 1b), R_{NBW}-H_2, R_{NBW}-CO_2 and R_{NBW}-Air achieved 225, 195 and 192 mL/g-VS, respectively, increasing by 39.3%, 20.8% and 19.1% in comparison to $R_{Control}$. The cumulative methane production from R_{NBW}-H_2 was significantly higher than that from $R_{Control}$.

The above results indicated that NBW addition promoted the production of biogas and methane from the AD of SM. This might be due to the addition of NBW promoted the hydrolysis of organic solids and shortened the lag period, which was beneficial to the subsequent acidification stage and methanogenesis stage. The enhancement in methane yield might also be due to the addition of NBW enhanced the activity of methanogens enzyme activity in the anaerobic digestion reactor. Improvement in methane production was also likely to be the result of the combination of these above factors. Therefore, the mechanisms involved in AD of SM under NBW addition requires further analysis.

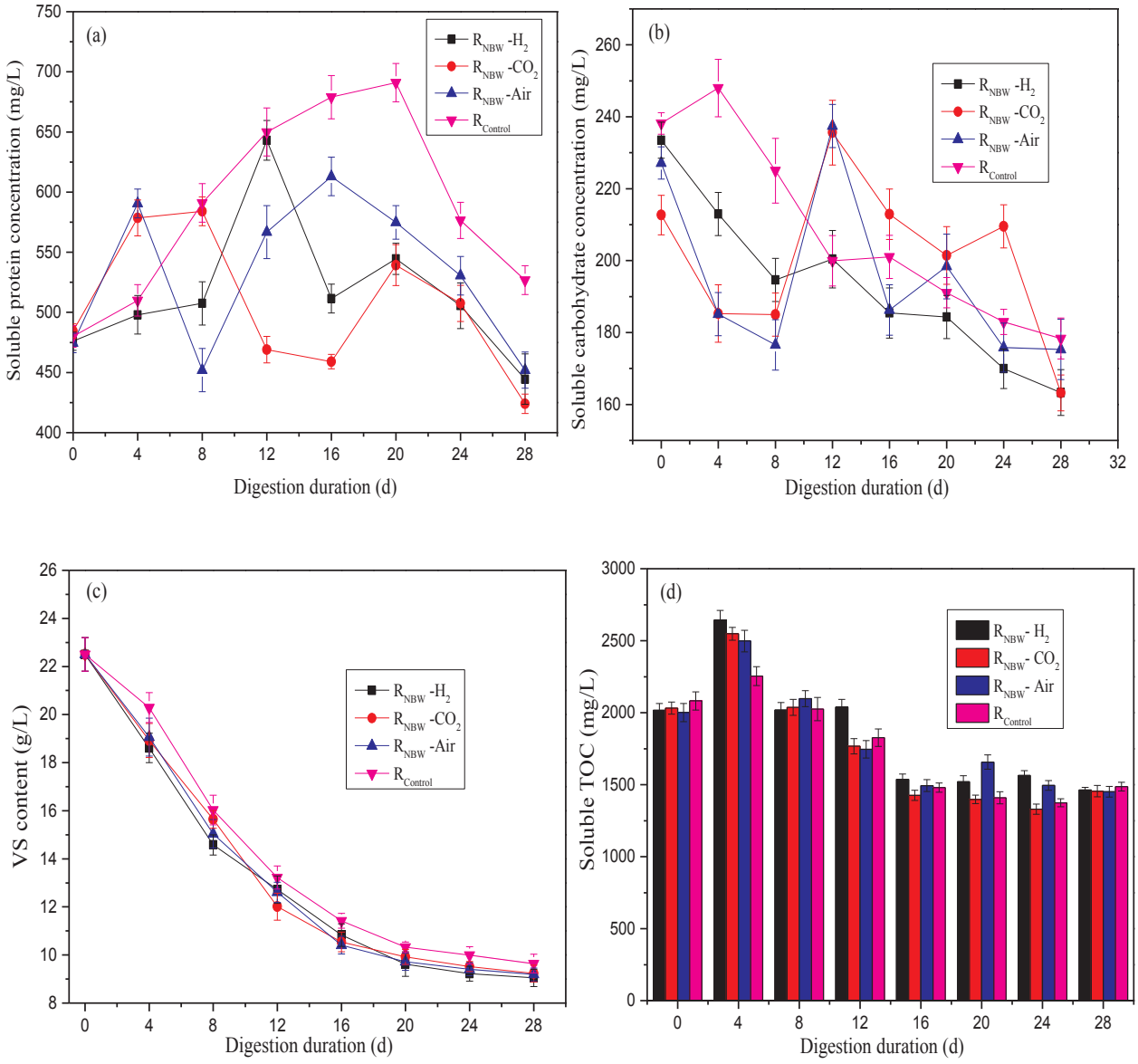


Fig.2. (a) Soluble protein concentration during AD process. (b) Soluble carbohydrate concentration during AD process. (c) VS content variation during AD process. (d) Soluble TOC variation during AD process.

3.2. Soluble organics

Successful AD process must be able to effectively degrade complex substrate into methane, while various compounds may inhibit or promote AD process (Li et al., 2016). Proteins, carbohydrates and lipids in organics contribute the most to methane production (Niu et al., 2014). The content of lipids in SM is relatively low, < 1% COD (Lin et al., 2015). Therefore, in this study, soluble proteins and carbohydrates were selected as key research subjects. It could be found from Fig. 2a that the soluble proteins in R_{NBW} decreased rapidly in the later stages of AD. However, the soluble proteins in $R_{Control}$ increased and reached a maximum of about 720 mg/L. Fig. 2b shows the variation of soluble carbohydrates. The trend of soluble carbohydrates was different from that of the soluble proteins. From the beginning to day 4, the soluble carbohydrates in R_{NBW} decreased rapidly, while that in $R_{Control}$ slowly increased and reached its maximum (248 mg/L) on day 4. As the AD progressed, the concentrations of soluble carbohydrates in $R_{Control}$ and R_{NBW} showed a downward trend.

At the beginning of the AD process, the VS contents in $R_{Control}$ and R_{NBW} were the same (Fig. 2c). Both $R_{Control}$ and R_{NBW} showed a trend of rapid decline and then slowly decline. However, the reduction rates of VS in R_{NBW} on day 4 (15.3–17.3%) and day 8 (30.5–35.2%) were faster than those in $R_{Control}$ (9.8% on day 4, and 28.8% on day 8), which indicated that the hydrolysis step in R_{NBW} might be enhanced in comparison to that in $R_{Control}$. The changes of soluble organics were the results from the balance between hydrolysis and consumption. From the results of soluble proteins and carbohydrates in this study, it could be concluded that the soluble organics in R_{NBW} were consumed faster than those in $R_{Control}$ at the early stage of AD. Considering the changes of VS at the initial stage of AD, it could be concluded that NBW addition not only enhanced the hydrolysis rate, but also accelerated the consumption rate of soluble organics. The consumption of soluble organics such as soluble proteins and carbohydrates can produce VFAs (Wang et al., 2014). Therefore, under NBW addition, the acidogenesis stage could be promoted, too. After 16 days'AD, the VS reduction rate slowed down in both $R_{Control}$ and R_{NBW}. At the end of AD tests, there was no statistically significant difference in VS reduction rates between $R_{Control}$ and R_{NBW} (58.9–59.8% in R_{NBW} and 57.2% in $R_{Control}$, $p > 0.05$). By analyzing the final VS content, it was clear that NBW addition could not significantly enhance VS reduction. These results indicated that the yield

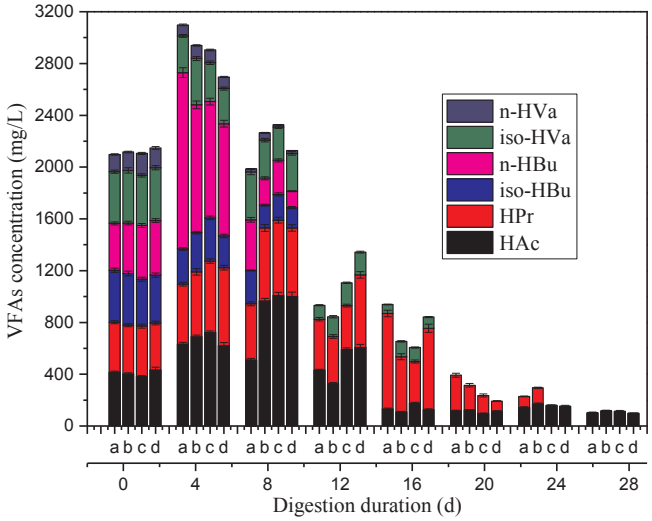


Fig 3. VFAs concentration variation of $R_{Control}$ and R_{NBW} during the whole AD process. The alphabets under X-axis areas: a-R_{NBW}-H_2, b-R_{NBW}-CO_2, c-R_{NBW}-Air, d-$R_{Control}$.

of CH_4 per unit VS removed in R_{NBW} was higher than that in $R_{Control}$. R_{NBW} was more efficient in using SM for methane production during AD.

Fig. 2d illustrates the changes of sTOC in SM during 28 days' AD with or without NBW addition. Obviously, the sTOC firstly increased and then decreased. At the beginning of the AD tests, the sTOC concentrations in R_{NBW} were higher than that in $R_{Control}$. As the AD progressed, the sTOC in R_{NBW} decreased rapidly, which was basically consistent with that in $R_{Control}$ during the late stage of AD. The maximum sTOC values in R_{NBW} were 10.8–17.2% higher than that in $R_{Control}$. According to the above analysis, the consumption rates of soluble organics in R_{NBW} were higher than $R_{Control}$. Therefore, at the initial stage of AD process, a higher sTOC concentration in R_{NBW} than in $R_{Control}$ also indicated that NBW addition enhanced the hydrolysis step.

VFAs include acetate (HAc), propionate (HPr), n-butyrate (n-HBu), iso-butyrate (iso-HBu), n-valerate (n-HVa), and iso-valerate (iso-HVa), which are products from hydrolysis and acidogenesis, and the substrates for methanogenesis (Li et al., 2015). VFAs are mainly produced in hydrolysis and acidogenesis stages of the AD process (Wang et al., 2014). It could be found from Fig. 3 that the initial VFAs concentrations in $R_{Control}$ and R_{NBW} were almost at the same level, about 2100 mg/L. When AD was progressed to day 4, the total concentrations of VFAs in R_{NBW} and $R_{Control}$ increased, and those in R_{NBW} (2900–3100 mg/L) were much higher than that in $R_{Control}$ (2700 mg/L). On day 8 and day 12, VFAs might be rapidly consumed by methanogens and no VFAs inhibition phenomenon was observed. Therefore, the higher VFAs concentrations in R_{NBW} at the early stage of AD might not be the result of less consumption. Instead, it was attributable to a much faster hydrolysis, which was consistent with the results of methane production. From day 4 to day 12, the methane production from R_{NBW} was higher than that of $R_{Control}$ by 25–53 mL/g-VS. Namely, NBW addition enhanced the hydrolysis and acidification stages in the AD of SM. This observation was also consistent with the results from the changes of VS and soluble proteins/carbohydrates.

3.3. Effects of NBW on related enzymes in AD process

The fluorescent properties of coenzyme F_{420} have been used to identify individual morphological types, colonies and growth rates of methanogens (Reynolds and Colleran, 1987). It can be seen from the Fig. 4a that the coenzyme F_{420} contents in $R_{Control}$ and R_{NBW} increased steadily during the first 20 days, and then showed a downward trend.

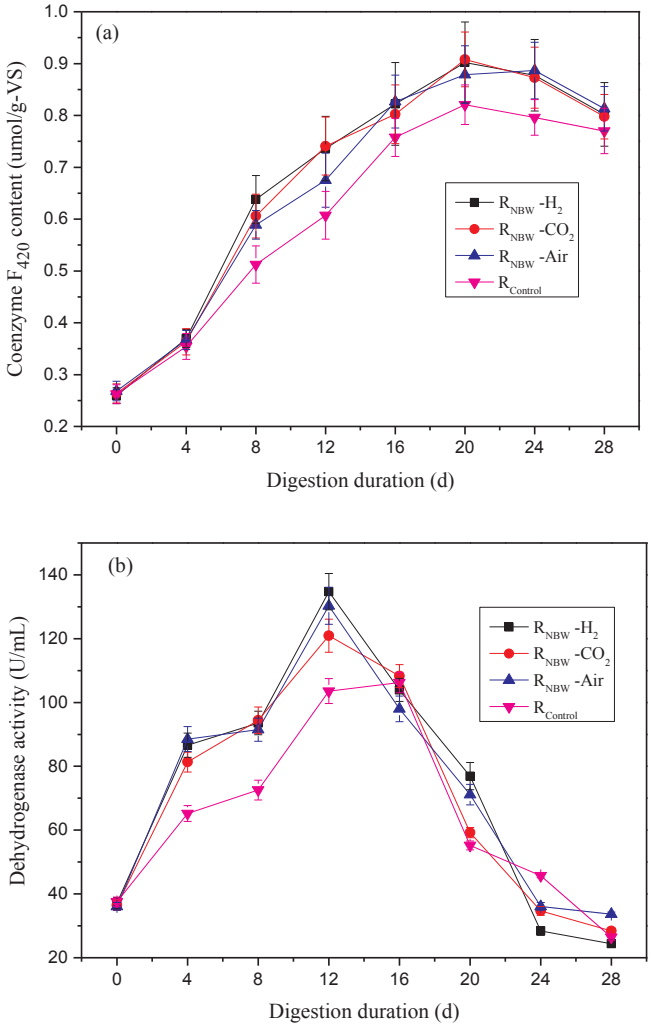


Fig.4. (a) Coenzyme F_{420} content variation during the AD process. (b) Dehydrogenase activity variation during the AD process.

The coenzyme F_{420} contents in R_{NBW} were higher than $R_{Control}$ but the difference was not so statistically significant ($p > 0.05$). Moreover, the difference among the three types of R_{NBW} was also not obvious ($p > 0.05$).

Dehydrogenase has the function of enzymatic dehydrogenation of organic substances, which can reflect the number of active microorganisms and the degradation activity of organic substances. As seen in Fig. 4b, the dehydrogenase activity in R_{NBW} was higher than that in $R_{Control}$ during the first 12 days of the AD process. This observation indicated that the activities of methanogen and dehydrogenase were enhanced by NBW addition. In addition, four enzymes (protease, α-glucosidase, ALP and ACP) were also measured to study the hydrolysis process. On day 4, the activities of ALP, ACP, α-glucosidase and protease in R_{NBW} were higher than those in $R_{Control}$, which also indicated that the hydrolysis process was enhanced due to the addition of NBW.

3.4. Changes in ETS and electric conductivity

ETS is associated with microbial respiratory activities in AD systems (Yin et al., 2005). Therefore, ETS can reflect the biological activities of microorganisms in the AD of SM. Fig. 5a shows the relative ETS in $R_{Control}$ and R_{NBW} throughout the AD process. The initial ETS in $R_{Control}$ was taken as the reference, which was 100. As seen from Fig. 5a, the relative ETS content in $R_{Control}$ and R_{NBW} were similar at the very beginning, indicating that the microbial activities in $R_{Control}$ and R_{NBW}

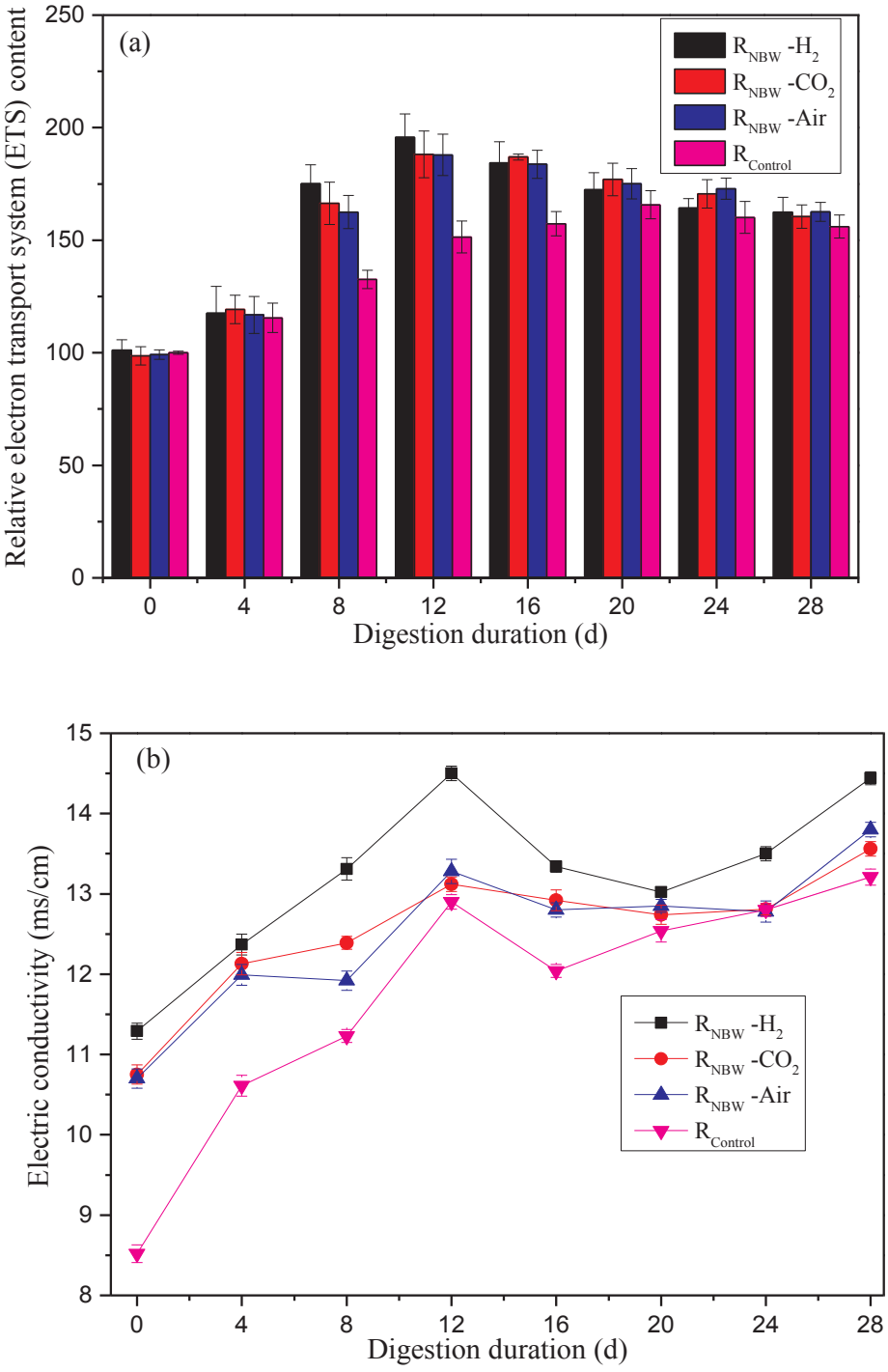


Fig.5. (a) Relative electron transport system variation during the AD process. (b) Electric conductivity variation during the AD process.

were basically the same at the initial stage. Before day 4, less increase was detected in the microbial activities in $R_{Control}$ and R_{NBW}, possibly due to the fact that the microorganisms had to adapt to the new environment during this period. However, from day 4 to day 12, the relative ETS content in R_{NBW} increased rapidly than that in $R_{Control}$. Fig. 5b demonstrated the changes of electric conductivity in these AD reactors during the whole process. It was found that the electric conductivities in R_{NBW} were higher than that in $R_{Control}$, among which R_{NBW}-H_2 showed significantly ($p = 0.043 < 0.05$) higher electric conductivity than $R_{Control}$.

3.5. Kinetic analysis

In this study, the modified Gompertz model and the first-order model were applied to better understand the kinetics of the AD process of SM under NBW addition. The relevant parameters are summarized in Table 3. Results showed that the modified Gompertz model well fit the experimental data from the AD process ($R^2 = 0.9978–0.9994$). Compared to $R_{Control}$, R_{NBW} showed relatively higher methane productions at significant difference $p < 0.0083$, which could be also proved by the higher P_{max} and R_{max} value. In addition, the higher first-order hydrolysis rate constant (k_H) of R_{NBW} also indicated that the addition of NBW could promote the hydrolysis stage of AD, which was also consistent

Table 3
Kinetic analysis of cumulative methane production from AD of SM with and without adding NBW.

Models	H_2-NBW	CO_2-NBW	Air-NBW	$R_{Control}$
Modified Gompertz				
R^2	0.9994	0.9986	0.9993	0.9978
λ (d)	3.15 ± 0.11	3.15 ± 0.17	3.40 ± 0.12	3.24 ± 0.22
Pmax (mL CH_4/g-VS)	235 ± 1	197 ± 2	186 ± 1	164 ± 2
Rmax (mL CH_4/g-VS·d)	19.2 ± 0.4	16.1 ± 0.5	15.2 ± 0.3	12.2 ± 0.4
First-order				
R^2	0.9544	0.9552	0.9511	0.9638
*P*max (mL CH4/g-VS)	346 ± 59	293 ± 51	288 ± 58	251 ± 43
k_H (d^{-1})	0.05 ± 0.01	0.05 ± 0.01	0.05 ± 0.01	0.04 ± 0.01
*P*max from the AD tests (mL CH4/g-VS)	225 ± 6 [a]	195 ± 3[b]	193 ± 2[b]	162 ± 4[c]

Data in the table are expressed as Means ± SD. The different letters (a, b, c) indicate significant difference at 0.83% level ($p < 0.0083$) using Bonferoni method for multiple comparison analysis.

with the analysis results of soluble organics. T_{90} was the time required to obtain 90% cumulative methane, which was the technically effective digestion time; and T_{ef} was calculated by subtracting the lag period from T_{90} (Kafle et al., 2014). The T_{90} for R_{NBW} was 0.77–1.48 d shorter than that for $R_{Control}$. The T_{ef} for $R_{Control}$ was 0.93–1.39 d shorter than that for $R_{Control}$. These results suggested that NBW addition could not only promote the production of methane but also enhance the hydrolysis of SM.

3.6. Mechanisms involved in AD of SM under NBW addition

In this work, multiple parallel experiments demonstrated that NBW addition could increase CH_4 yield in the AD of SW. In terms of soluble organics, the addition of NBW accelerated the VS reduction at the initial stage of AD, indicating that the hydrolysis of SM was enhanced due to NBW addition. Moreover, by analyzing the changes of VFAs and soluble proteins/carbohydrates, it can be seen that NBW addition might enhance the hydrolysis and acidification stages in the AD process. However, the addition of NBW did not significantly increase the VS reduction rates when the entire AD process was considered. In addition, NBW addition did not change the compositions of the inoculum and raw SM, nor did it change carbon to nitrogen ratio.

In order to clarify the mechanisms of NBW addition for improving AD of SM, the T_2 and zeta potential were analyzed in this work. As shown in Fig. 6, the T_2 values in the three types of NBW (2803.49, 2873.21 and 3012.71 ms) were longer than that in DW (2459.21 ms), and the maximum value of T_2 was detected in H_2-NBW. The relatively long T_2 value could be contributed by the weakening of the hydrogen-bonding network in water, which increases the mobility of the water molecules (Liu et al., 2013). That is, the hydrogen-bonding network in R_{NBW} was weaker and the mobility of the water molecules in R_{NBW} was stronger than that in $R_{Control}$. As known, zeta potential plays a key role in bubble stability (Parmar and Majumder, 2013). Colloids with high zeta potential contribute to electrical stability and avoid coalescence due to repulsion. As it can be expected, due to the addition of NBW, the absolute value of zeta potential could be increased. In this study, since NBW has a higher absolute value of zeta potential (16.5–21.1 mV) than DW (1.7 mV), addition of NBW may avoid the formation of aggregates and maintain the long-term electrical stability (Fig. 6). When NBs reach the critical size, these bubbles will violently collapse during their growth. As stated, cavitation collapse can not only produce liquid–vapor interface high pressure, strong local heating, high shear and turbulence, but also free radicals (Bougrier et al., 2006).

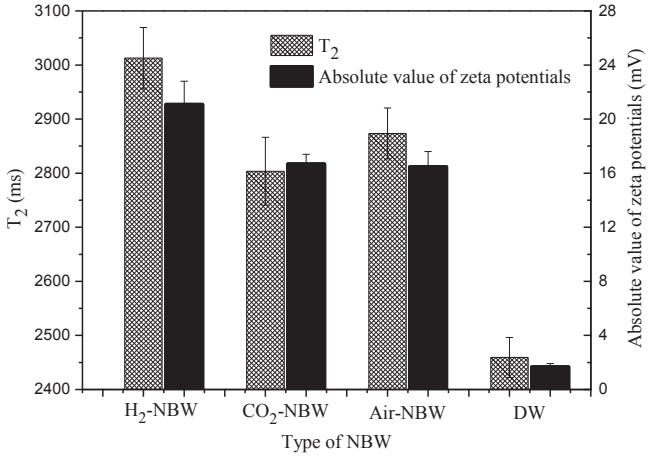


Fig.6. T_2 values and absolute value of zeta potentials of NBW and DW.

The number of bubble particles was reported to be still around 1×10^8 particles/mL when NBW was stored for 14 days (Wang et al., 2019). Therefore, when NBW was added to the AD of SM, NBW might play its role in the AD process for a relatively long time. When NBW with higher T_2 value and zeta potential was added into the AD of SM, the enzyme activities were improved. From the activities of the four hydrolases (ALP, ACP, α-glucosidase and protease), it could be concluded that addition of NBW increased the hydrolysis rates. This result was consistent with the results from the kinetic model analysis. Therefore, NBW addition promoted hydrolysis stage during AD of SM. The increase in hydrolysis rate facilitates subsequent acidification stage and methanogenesis stage, which is beneficial to increase methane production and improve AD performance.

In addition, the changes of Coenzyme F_{420} contents indicated that NBW addition increased the activity of methanogens. On the other hand, the activity of dehydrogenase was also enhanced under NBW addition. The enhanced enzyme activities might be accompanied by the increased activities of microbial populations such as methanogens. The changes in ETS and electric conductivity in R_{NBW} could further confirm the improved activities of microbial population after NBW supplementation, resulting in faster utilization of soluble organics and then increased methane production.

Therefore, NBW with higher T_2 value and the absolute value of zeta potential made the solution in the AD reactor have higher mobility of water molecules, weaker hydrogen-bonding network and higher electric conductivity, which could enhance the activities of methanogens and related enzymes, thereby further increasing methane production and improving AD performance. Results from this study suggested that NBW supplementation might be a promising approach for AD of SM. Future works are still demanding on the in-depth mechanisms of NBW in the AD process, characteristics of the digestate, and the impact of its application in farmland.

4. Conclusions

(1) Biogas and methane production were enhanced respectively by 16.1%-31.5% and 19.1%-39.3% under the addition of NBW.
(2) NBW addition was found to enhance the hydrolysis and acidification stages in the AD of SM.
(3) Coenzyme F_{420} and dehydrogenase were increased under NBW addition, indicating the activities of methanogen and dehydrogenase were enhanced. ETS and electric conductivity were also increased, signaling the enhanced activity of microorganisms.
(4) The increases in T_2 and the absolute value of zeta potential reflected the weaker hydrogen-bonding network and the greater mobility of water molecules.

CRediT authorship contribution statement

Yujie Fan: Conceptualization, Investigation, Validation, Visualization, Writing - original draft. **Zhongfang Lei:** Conceptualization, Supervision, Writing - review & editing. **Zitao Guo:** Methodology. **Weiwei Huang:** Writing - review & editing. **Di Wang:** Formal analysis, Software. **Xuezhi Wang:** Resources. **Zhenya Zhang:** Conceptualization, Supervision, Writing - review & editing, Project administration. **Kazuya Shimizu:** Conceptualization, Supervision, Writing - review & editing, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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